B-Splines

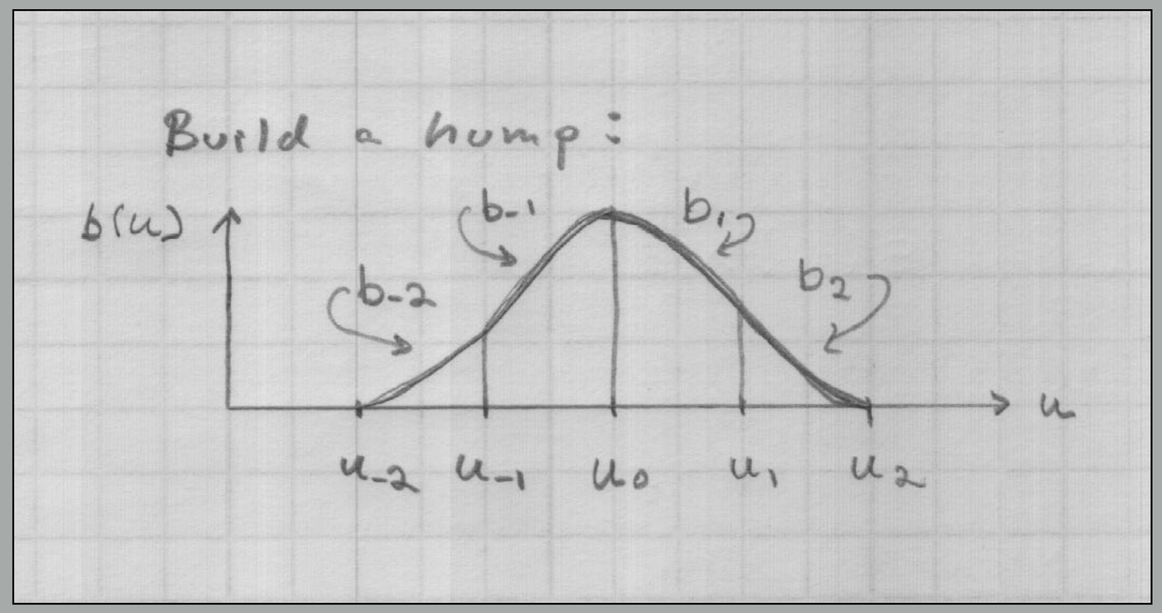
B-Splines

Goal: C^2 cubic curves with local support

- Give up interpolation
- Get convex hull property

Build basis by designing “hump” functions

B-Splines

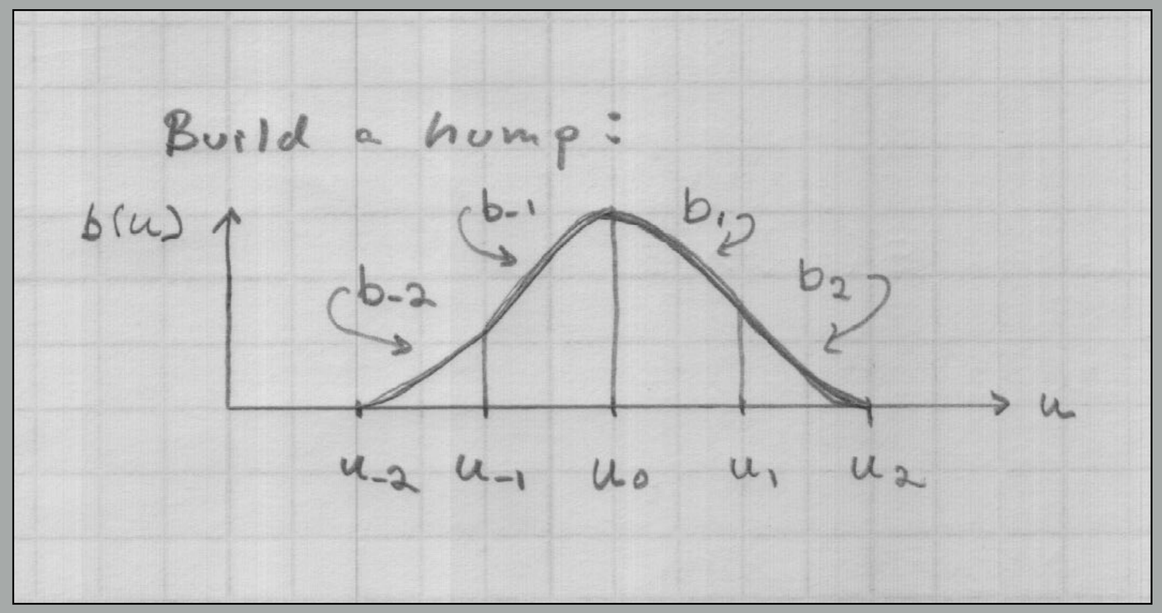


$$\mathbf{b}(u) = \begin{cases} \mathbf{b_{-2}}(u) & \text{if } u_{-2} \leq u < u_{-1} \\ \mathbf{b_{-1}}(u) & \text{if } u_{-1} \leq u < u_0 \\ \mathbf{b_{+1}}(u) & \text{if } u_0 \leq u < u_{+1} \\ \mathbf{b_{+2}}(u) & \text{if } u_{+1} \leq u \leq u_{+2} \end{cases}$$

$$b''_{-2}(u_{-2}) = b'_{-2}(u_{-2}) = b_{-2}(u_{-2}) = 0$$
$$b''_{+2}(u_{+2}) = b'_{+2}(u_{+2}) = b_{+2}(u_{+2}) = 0$$

$$b_{-2}(u_{-1}) = b_{-1}(u_{-1})$$
$$b_{-1}(u_0) = b_{+1}(u_0)$$
$$b_{+1}(u_{+1}) = b_{+2}(u_{+1})$$

B-Splines



$$\mathbf{b}(u) = \begin{cases} \mathbf{b_{-2}}(u) & \text{if } u_{-2} \leq u < u_{-1} \\ \mathbf{b_{-1}}(u) & \text{if } u_{-1} \leq u < u_0 \\ \mathbf{b_{+1}}(u) & \text{if } u_0 \leq u < u_{+1} \\ \mathbf{b_{+2}}(u) & \text{if } u_{+1} \leq u \leq u_{+2} \end{cases}$$

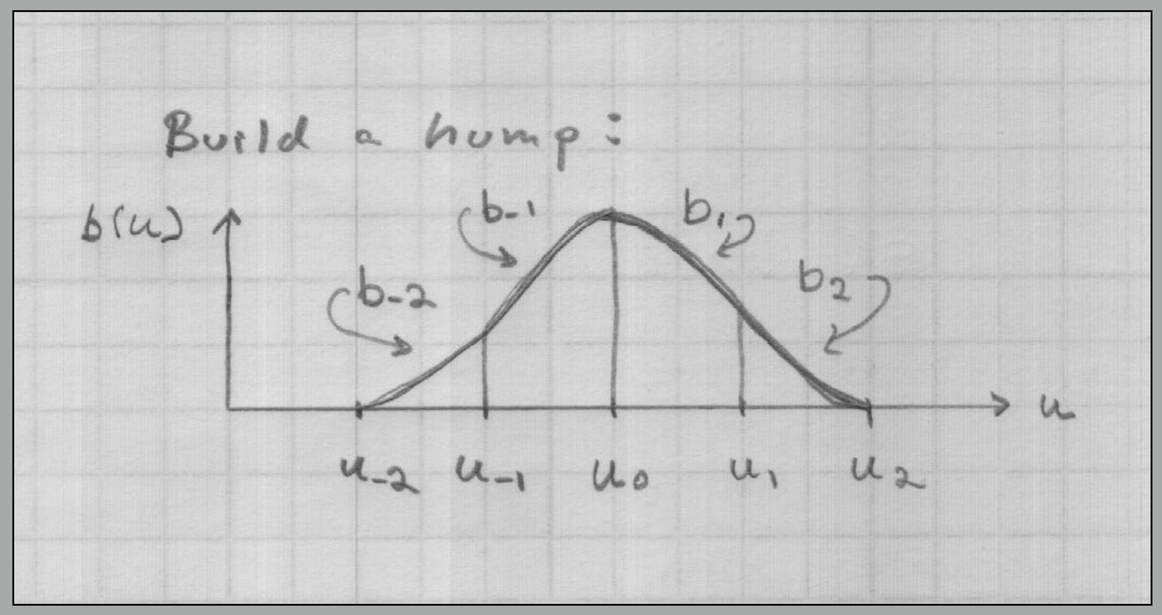
$$b''_{-2}(u_{-2}) = b'_{-2}(u_{-2}) = b_{-2}(u_{-2}) = 0 \quad \leftarrow 3 \text{ constraints}$$
$$b''_{+2}(u_{+2}) = b'_{+2}(u_{+2}) = b_{+2}(u_{+2}) = 0 \quad \leftarrow 3 \text{ constraints}$$

$$b_{-2}(u_{-1}) = b_{-1}(u_{-1})$$
$$b_{-1}(u_0) = b_{+1}(u_0) \quad \leftarrow$$
$$b_{+1}(u_{+1}) = b_{+2}(u_{+1})$$

Repeat for b' and b''
$3 \times 3 = 9$ constraints

Total 15 constraints ..... need one more

B-Splines



$$\mathbf{b}(u) = \begin{cases} \mathbf{b_{-2}}(u) & \text{if } u_{-2} \leq u < u_{-1} \\ \mathbf{b_{-1}}(u) & \text{if } u_{-1} \leq u < u_0 \\ \mathbf{b_{+1}}(u) & \text{if } u_0 \leq u < u_{+1} \\ \mathbf{b_{+2}}(u) & \text{if } u_{+1} \leq u \leq u_{+2} \end{cases}$$

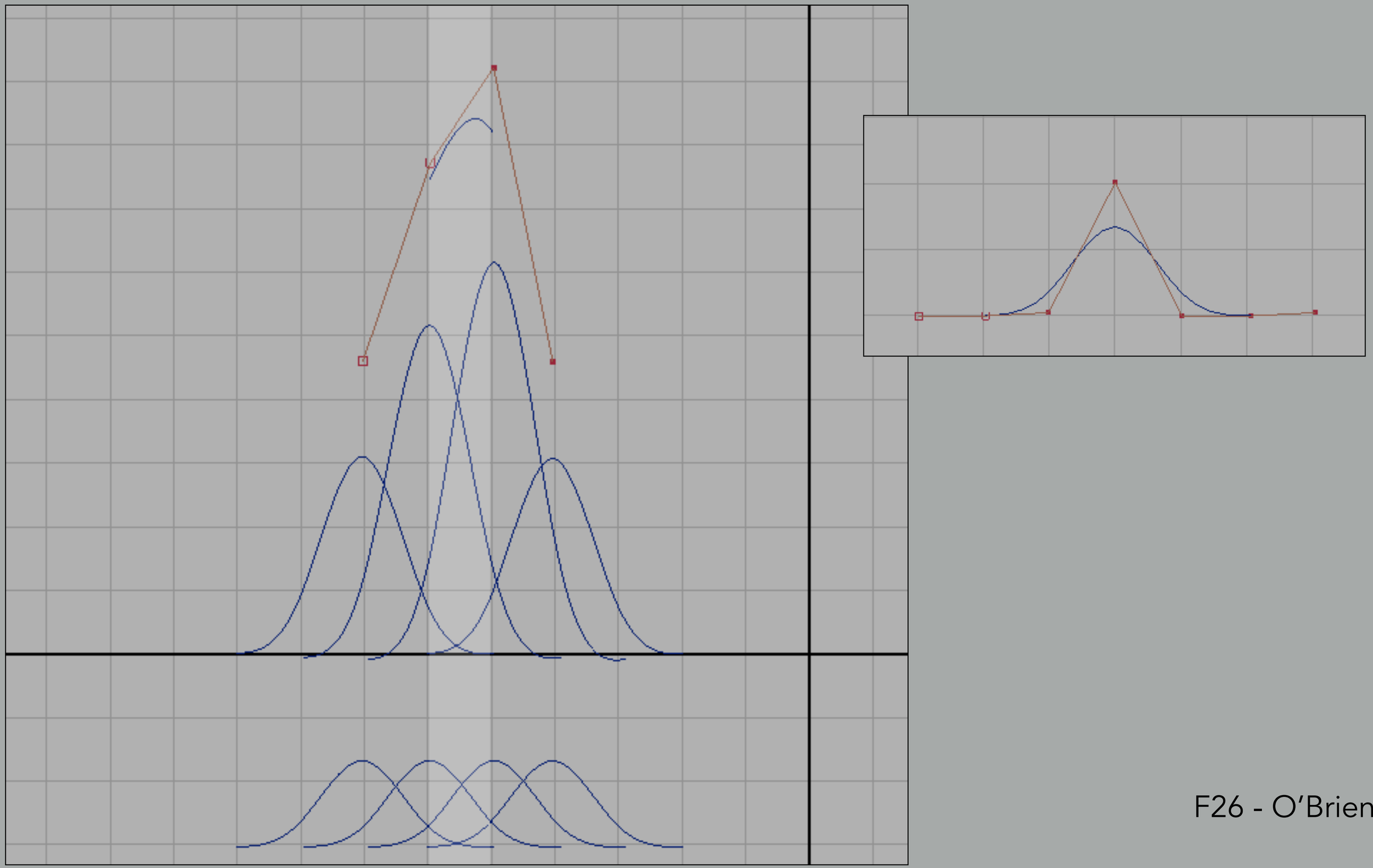
$$b''_{-2}(u_{-2}) = b'_{-2}(u_{-2}) = b_{-2}(u_{-2}) = 0 \quad \leftarrow 3 \text{ constraints}$$
$$b''_{+2}(u_{+2}) = b'_{+2}(u_{+2}) = b_{+2}(u_{+2}) = 0 \quad \leftarrow 3 \text{ constraints}$$

$$\left.\begin{aligned} b_{-2}(u_{-1}) &= b_{-1}(u_{-1}) \\ b_{-1}(u_0) &= b_{+1}(u_0) \\ b_{+1}(u_{+1}) &= b_{+2}(u_{+1}) \end{aligned}\right\} \leftarrow \left[\begin{array}{l} \text{Repeat for } b' \text{ and } b'' \\ 3\times 3 = 9 \text{ constraints} \end{array}\right.$$

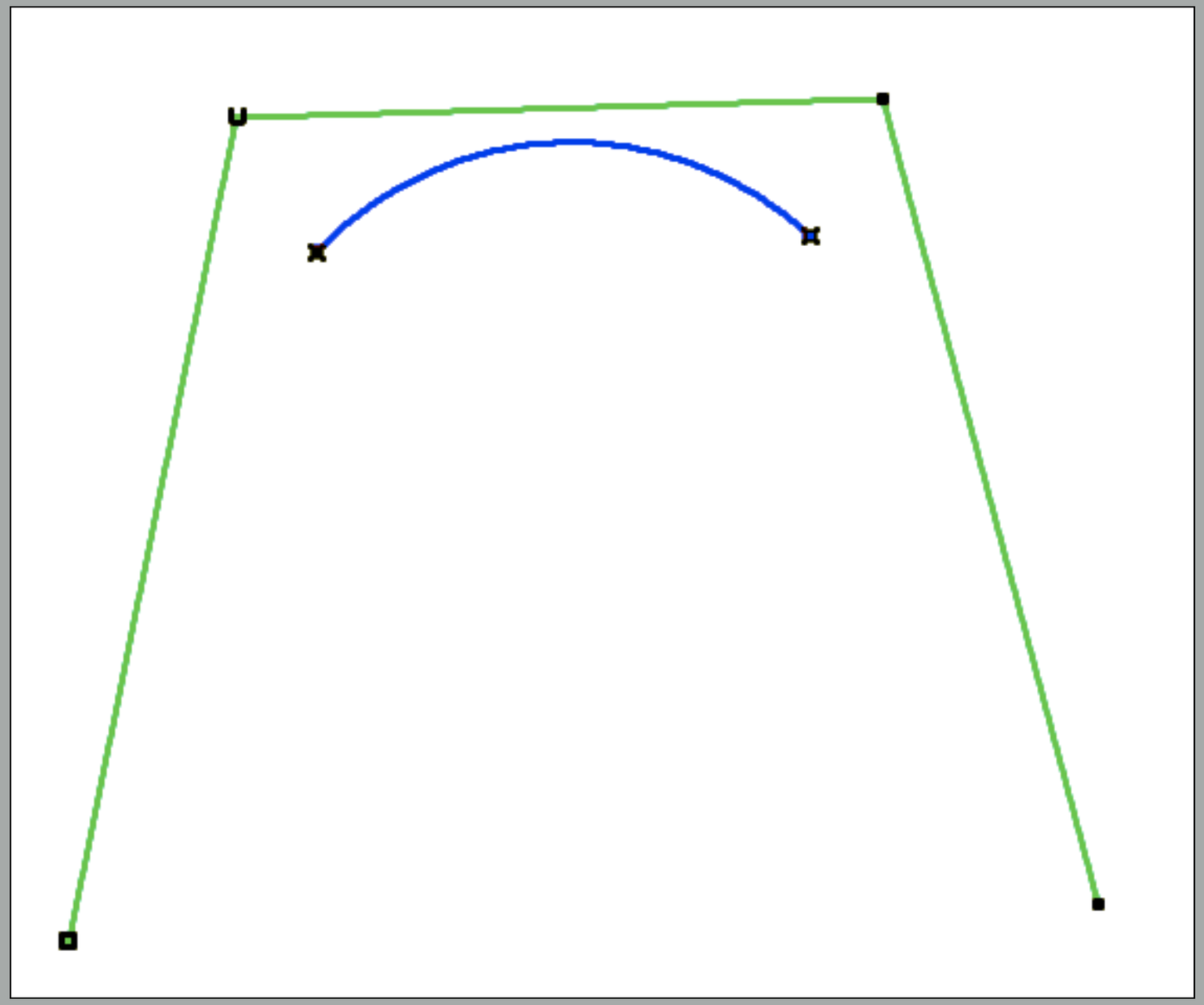
$$b_{-2}(u_{-2}) + b_{-1}(u_{-1}) + b_{+1}(u_0) + b_{+2}(u_{+1}) = 1 \quad \leftarrow 1 \text{ constraint (convex hull)}$$

Total 16 constraints

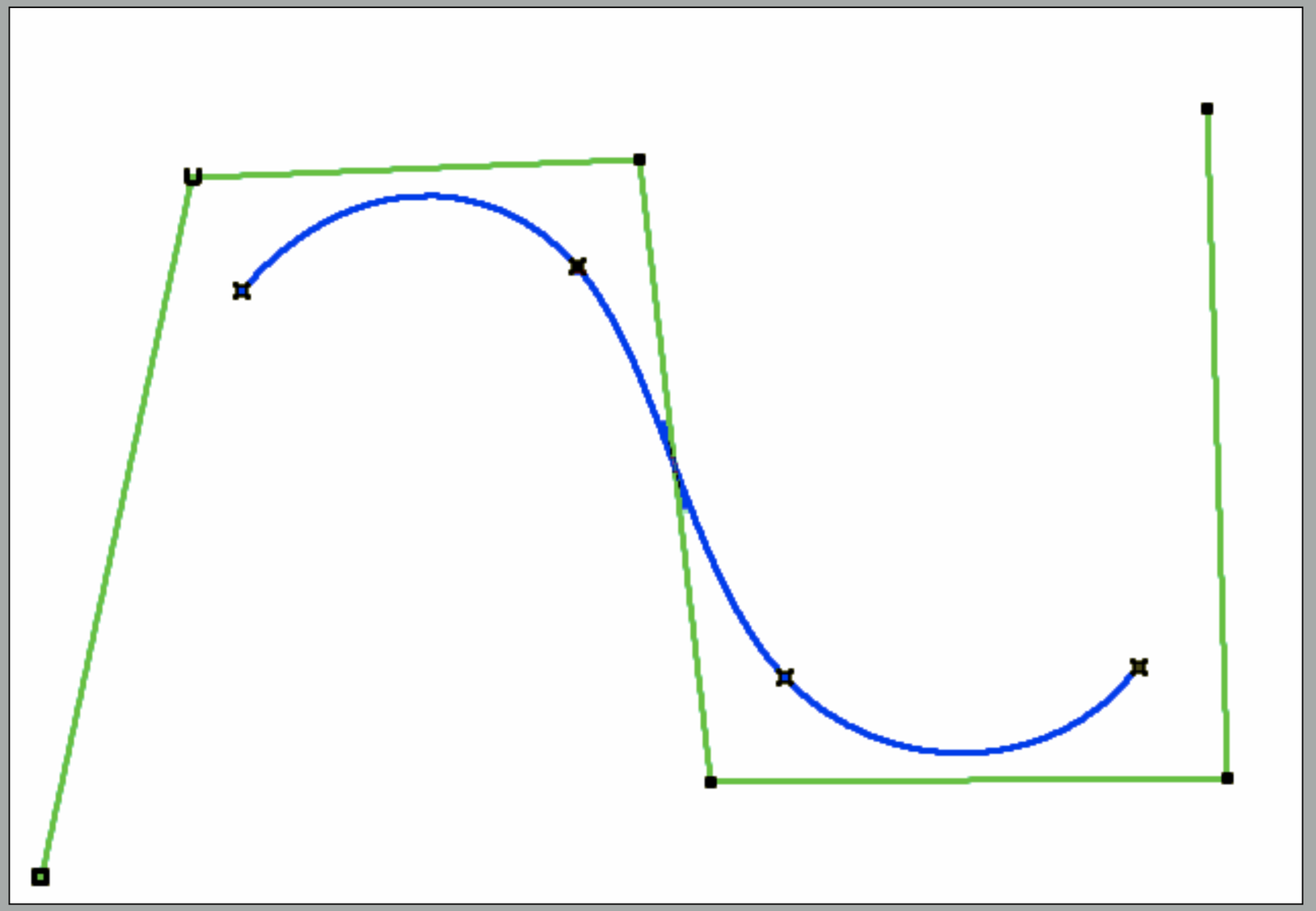
B-Splines



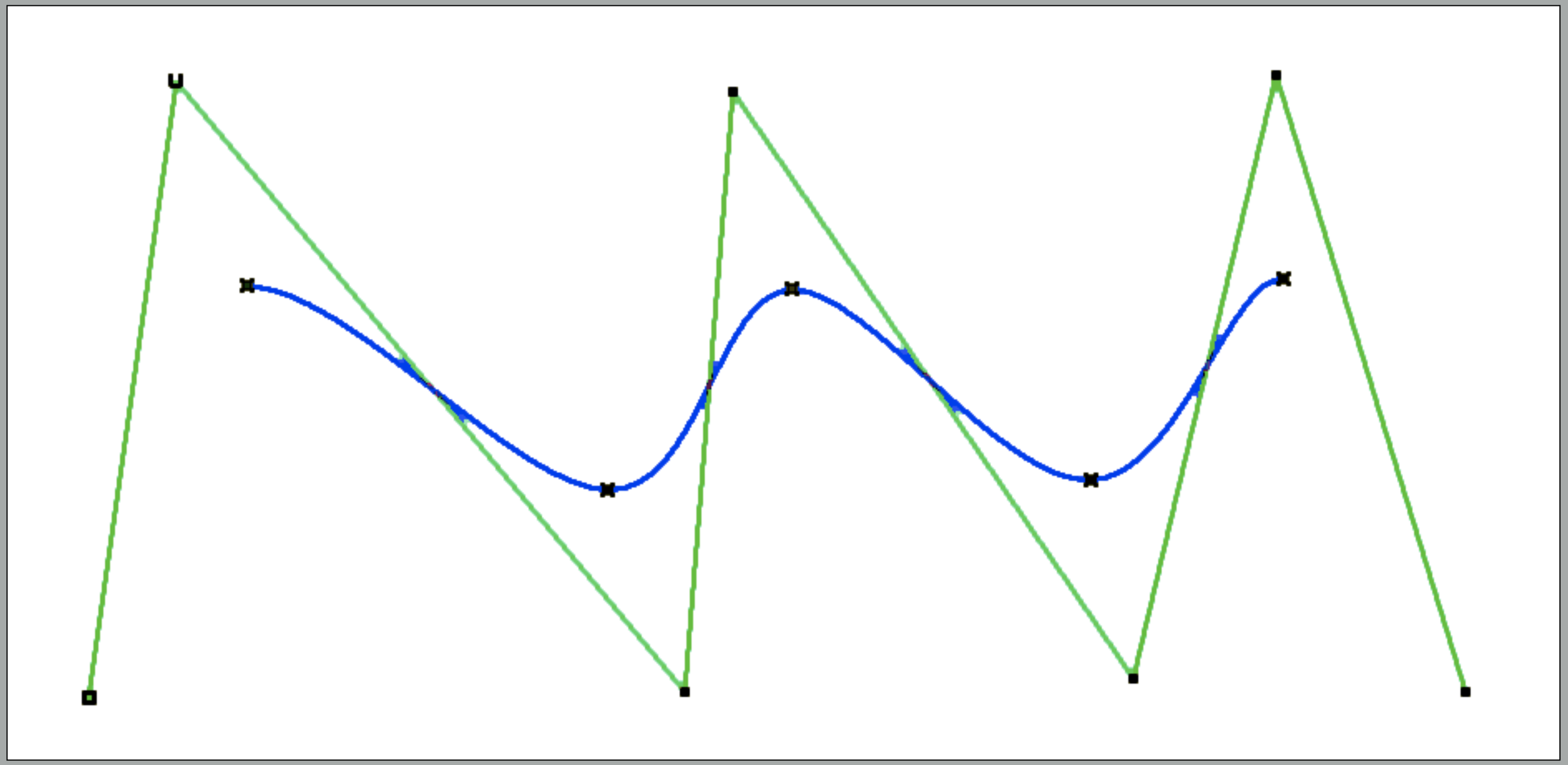
B-Splines



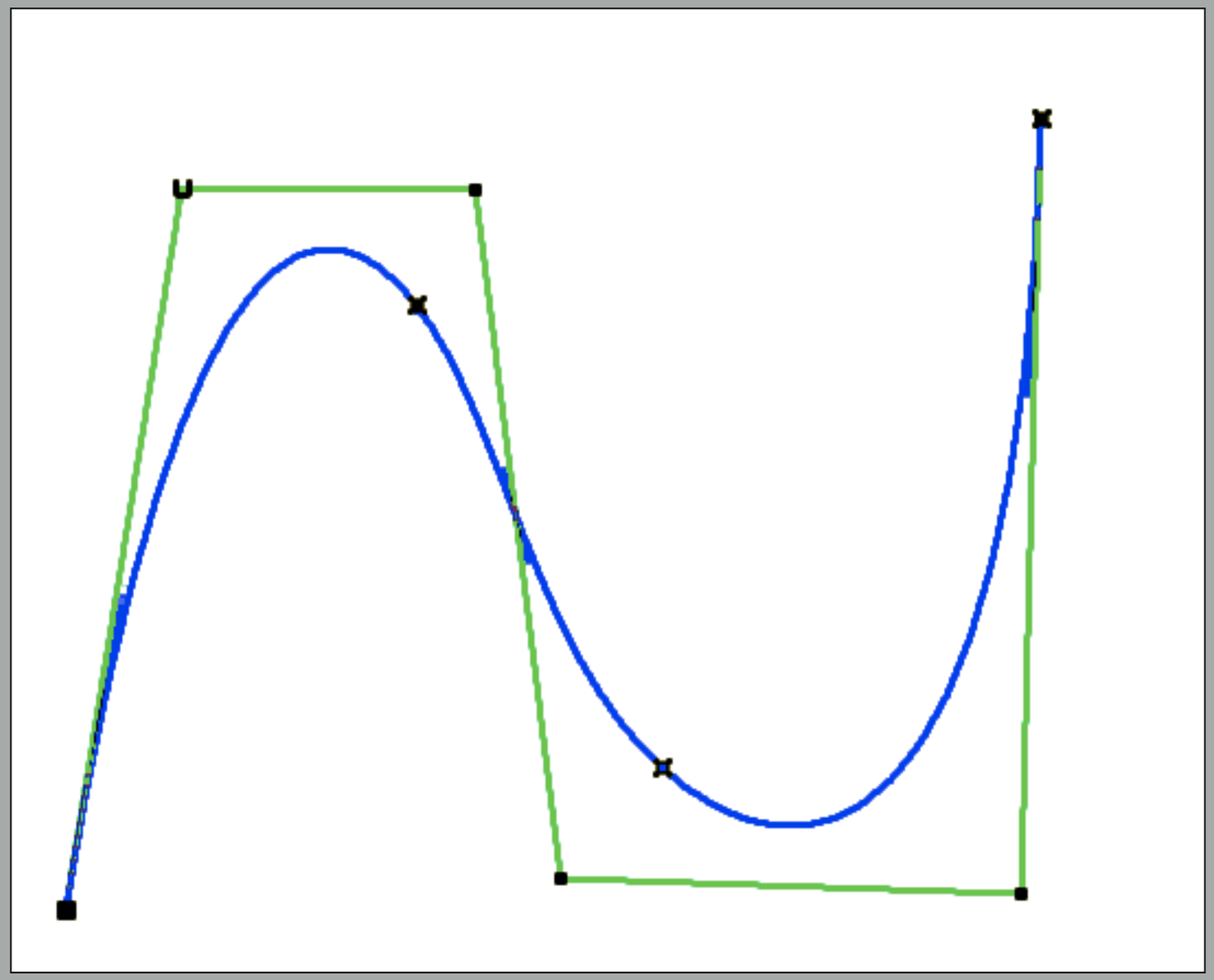
B-Splines



B-Splines



B-Splines



Example with end knots repeated

B-Splines

Build a curve w/ overlapping bumps

Continuity

- Inside bumps C^2
- Bumps "fade out" with C^2 continuity

Boundaries

- Circular
- Repeat end points
- Extra end points

B-Splines

- Notation
  - The basis functions are the $b_i(u)$
  - "Hump" functions are the concatenated function
    - Sometimes the humps are called basis... can be confusing
  - The u_i are the knot locations
  - The weights on the hump/basis functions are control points

B-Splines

- Similar construction method can give higher continuity with higher degree polynomials
- Repeating knots drops continuity
  - Limit as knots approach each other
- Still cubics, so conversion to other cubic basis is just a matrix multiplication

B-Splines

- Geometric construction
  - Due to Cox and de Boor
  - My own notation, beware if you compare w/ text

- Let hump centered on u_i be $N_{i,4}(u)$

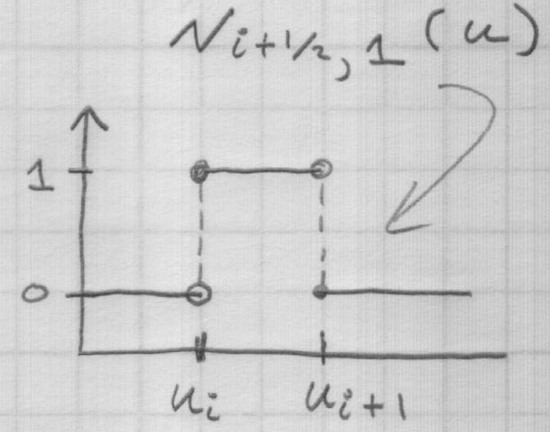
Cubic is order 4

$N_{i,k}(u)$ Is order k hump, centered at u_i

Note: i is integer if k is even
else $(i + 1/2)$ is integer

B-Splines

$$N_{i,1}(u) = \begin{cases} 1 & \text{if } u_{i-1/2} \le u < u_{i+1/2} \\ 0 & \text{else} \end{cases}$$

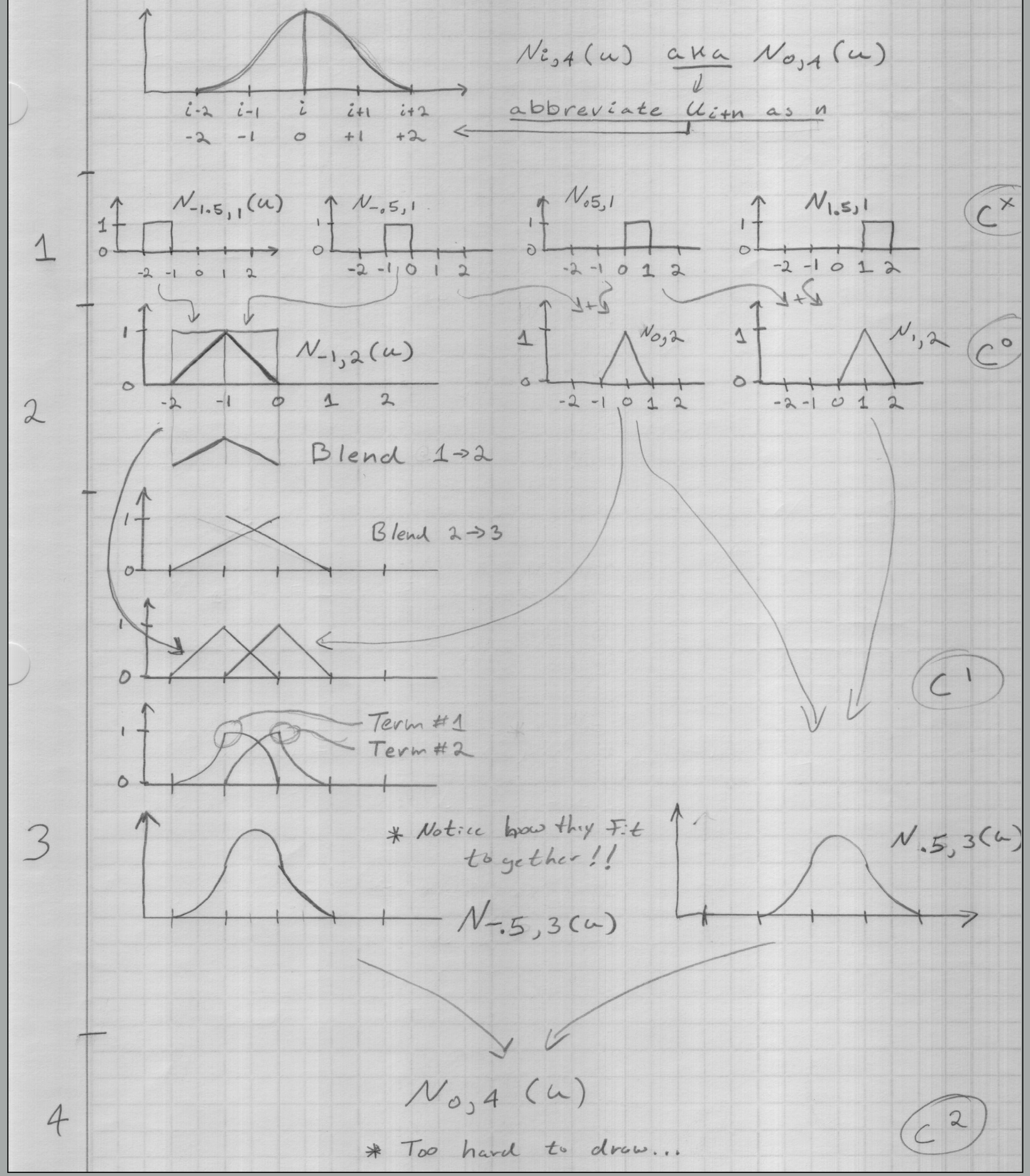


$$N_{i,k}(u) = \frac{(u - u_{i-k/2})\, N_{i-1/2,k-1}(u)}{u_{i+k/2-1} - u_{i-k/2}} + \frac{(u_{i+k/2} - u)\, N_{i+1/2,k-1}(u)}{u_{i+k/2} - u_{i-k/2+1}}$$

$k \ge 2$

"Term #1" "Term #2"

recursive defn.


$N_{i,4}(u)$ aka $N_{0,4}(u)$
abbreviate u_{i+n} as n
i-2 i-1 i i+1 i+2
-2 -1 0 +1 +2
1
$N_{-1.5,1}(u)$
$N_{-.5,1}$
$N_{.5,1}$
$N_{1.5,1}$
C^{-1}
2
$N_{-1,2}(u)$
$N_{0,2}$
$N_{1,2}$
C^0
Blend 1→2
Blend 2→3
C^1
Term #1
Term #2
3
* Notice how they fit together!!
$N_{-.5,3}(u)$
$N_{.5,3}(u)$
$N_{0,4}(u)$
* Too hard to draw...
C^2
4

NURBS

- **N**on**u**niform **R**ational **B**-**S**plines
  - Basically B-Splines using homogeneous coordinates
  - Transform under perspective projection
  - A bit of extra control

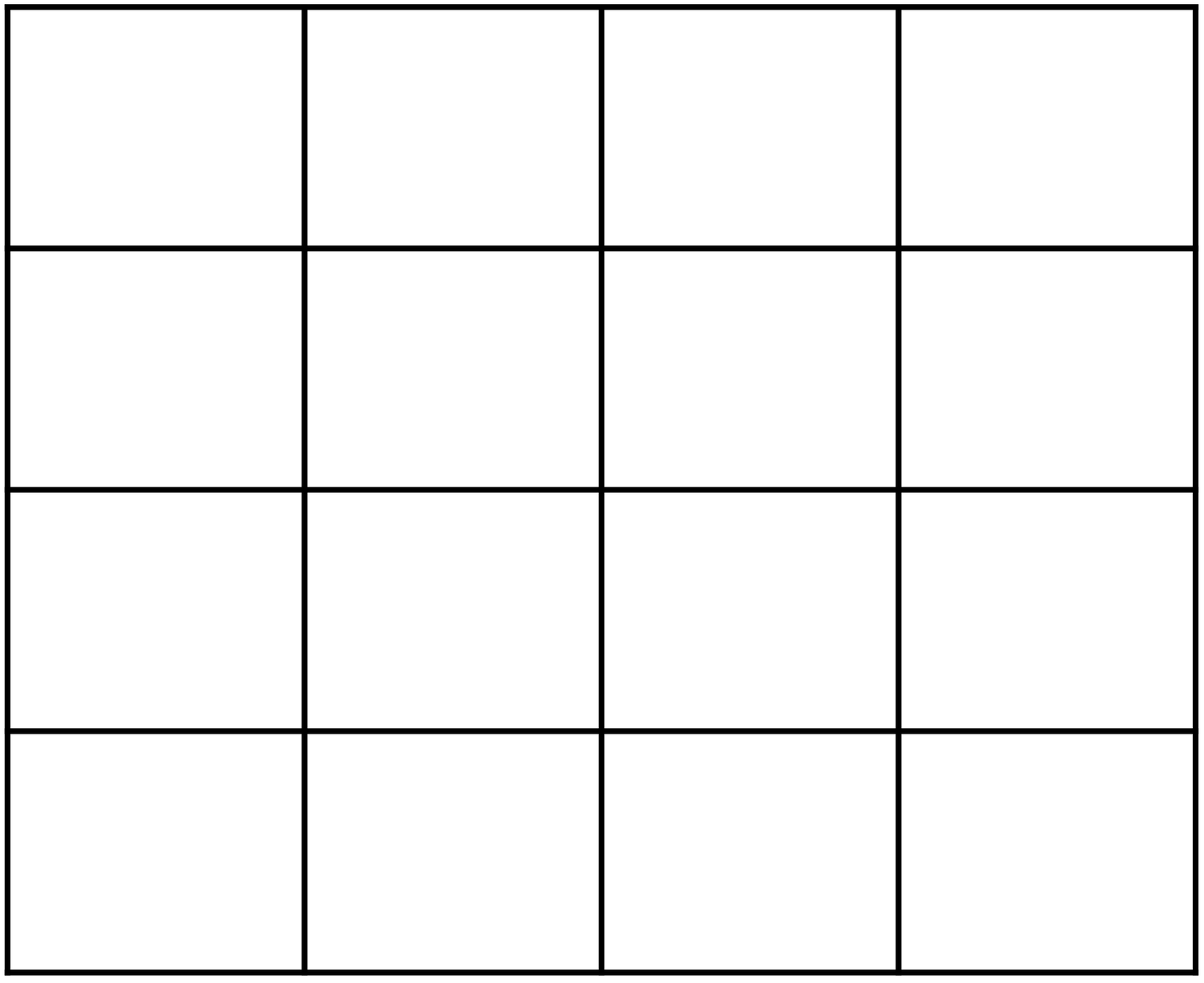
NURBS

$$\mathbf{p}_i = \begin{bmatrix} p_{ix} \\ p_{iy} \\ p_{iz} \\ p_{iw} \end{bmatrix} \qquad \mathbf{x}(u) = \frac{\sum_i \begin{bmatrix} p_{ix} \\ p_{iy} \\ p_{iz} \end{bmatrix} N_i(u)}{\sum_i p_{iw} N_i(u)}$$

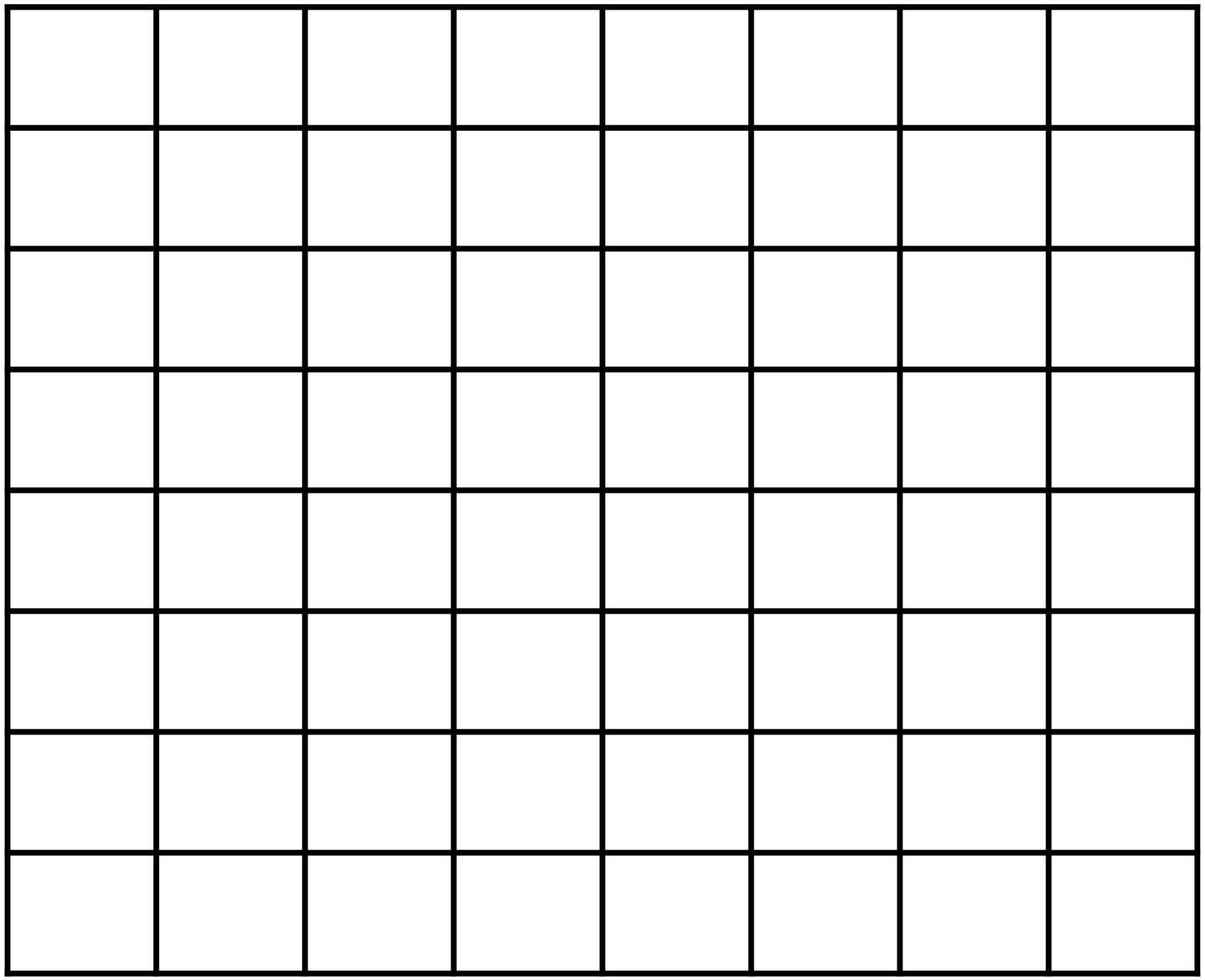
- Non-linear in the control points
- The p_{iw} are sometimes called "thumb weights" because increasing the value will pull curve toward corresponding control point.

Catmull-Clark Subdivision (Generalization of B-Splines)

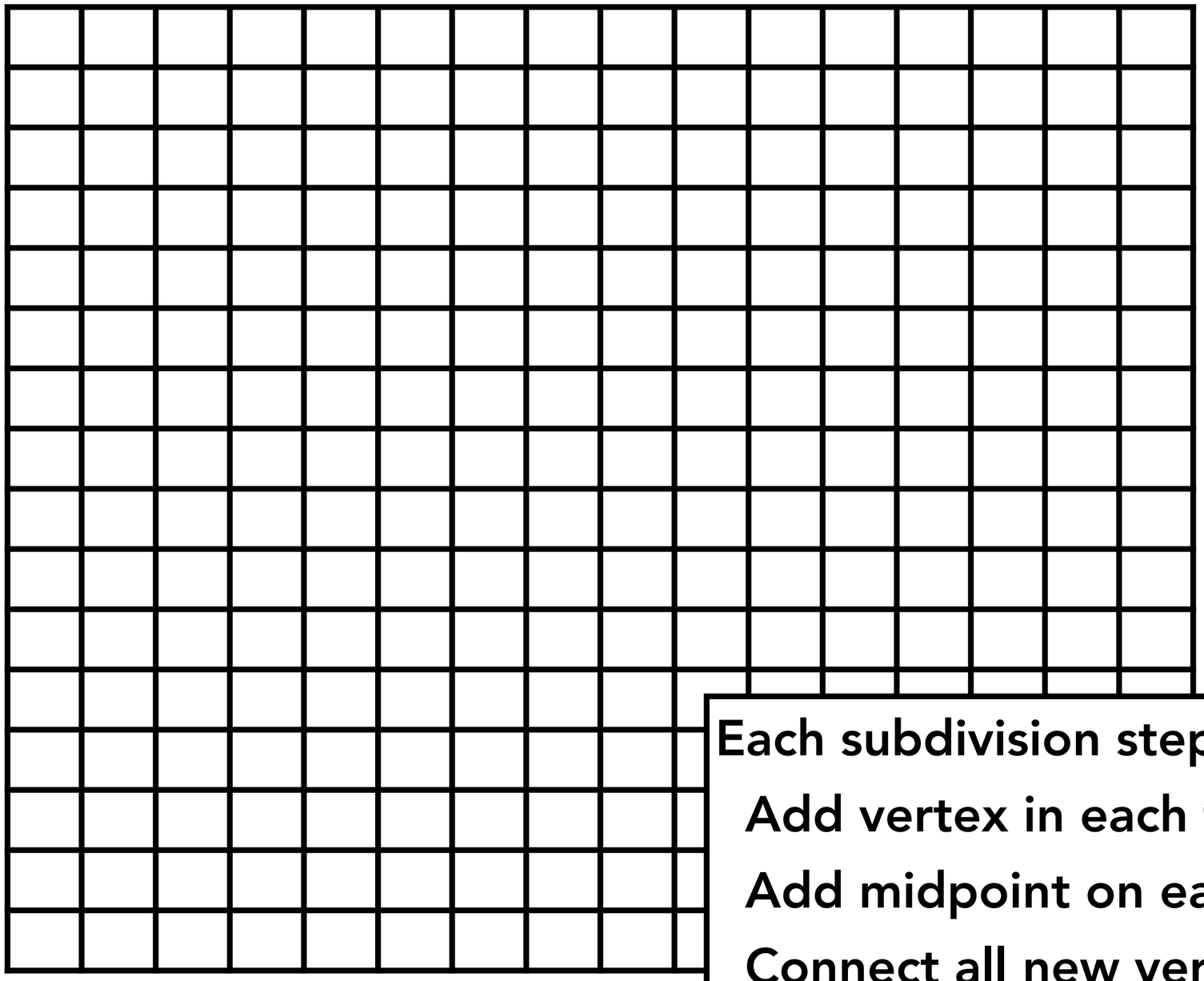
B-Spline Subdivision (Regular Quad Mesh)



B-Spline Subdivision (Regular Quad Mesh)



B-Spline Subdivision (Regular Quad Mesh)

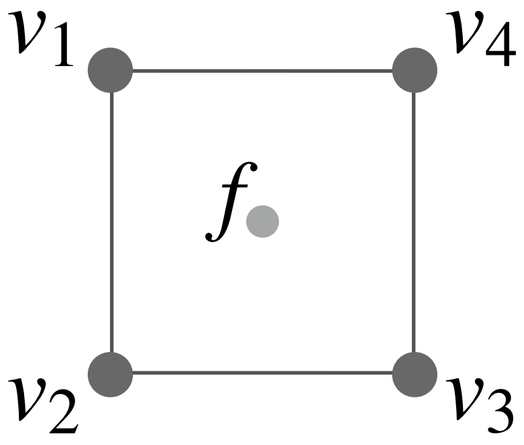


Each subdivision step:

- **Add vertex in each face**
- **Add midpoint on each edge**
- **Connect all new vertices**

B-Spline Vertex Update Rules (Quad Mesh)

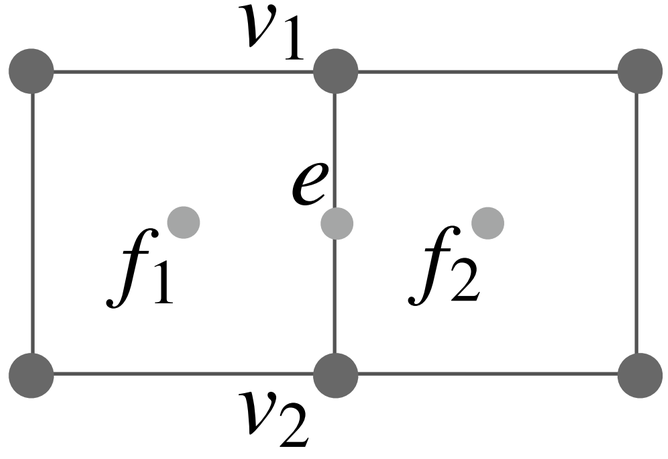
Face point

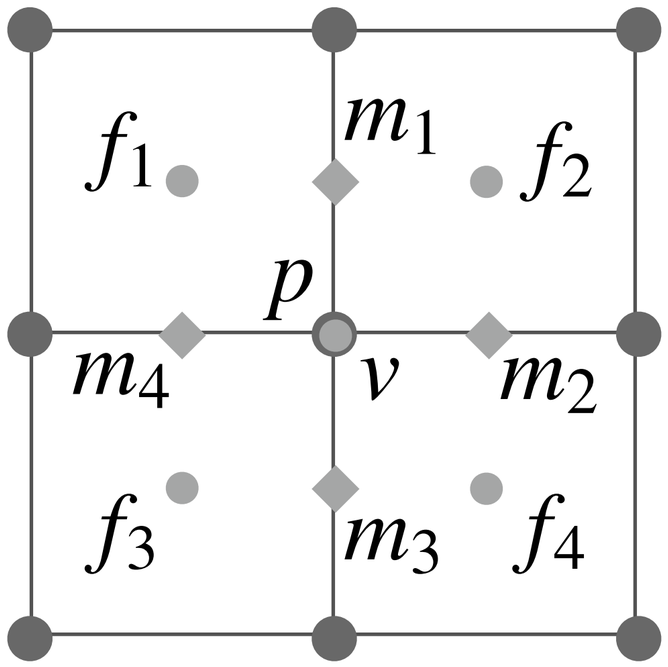


$$f = \frac{v_1 + v_2 + v_3 + v_4}{4}$$

$$e = \frac{v_1 + v_2 + f_1 + f_2}{4}$$

Edge point





Vertex point

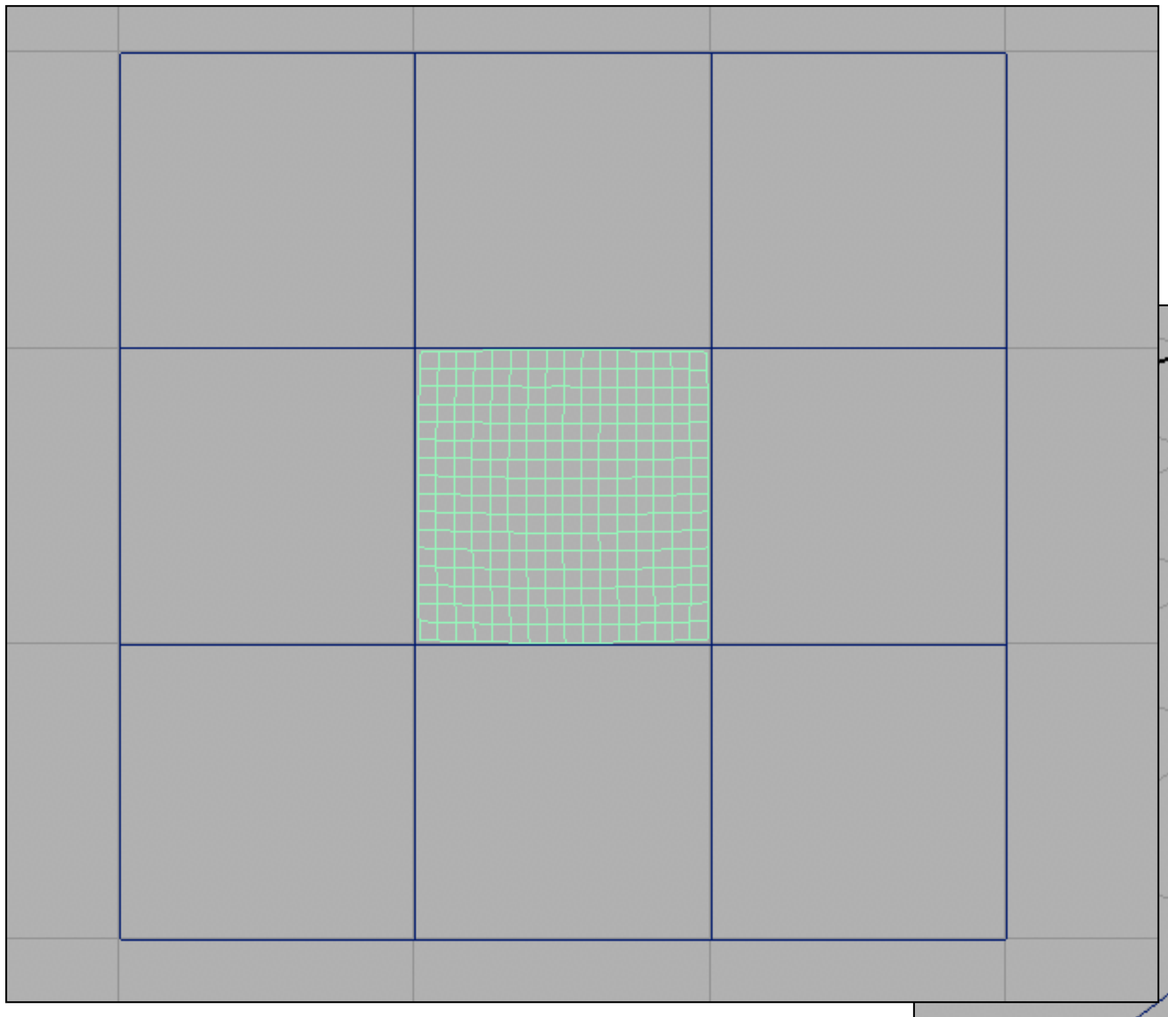
$$v = \frac{f_1 + f_2 + f_3 + f_4 + 2(m_1 + m_2 + m_3 + m_4) + 4p}{16}$$

m midpoint of edge, not "edge point"

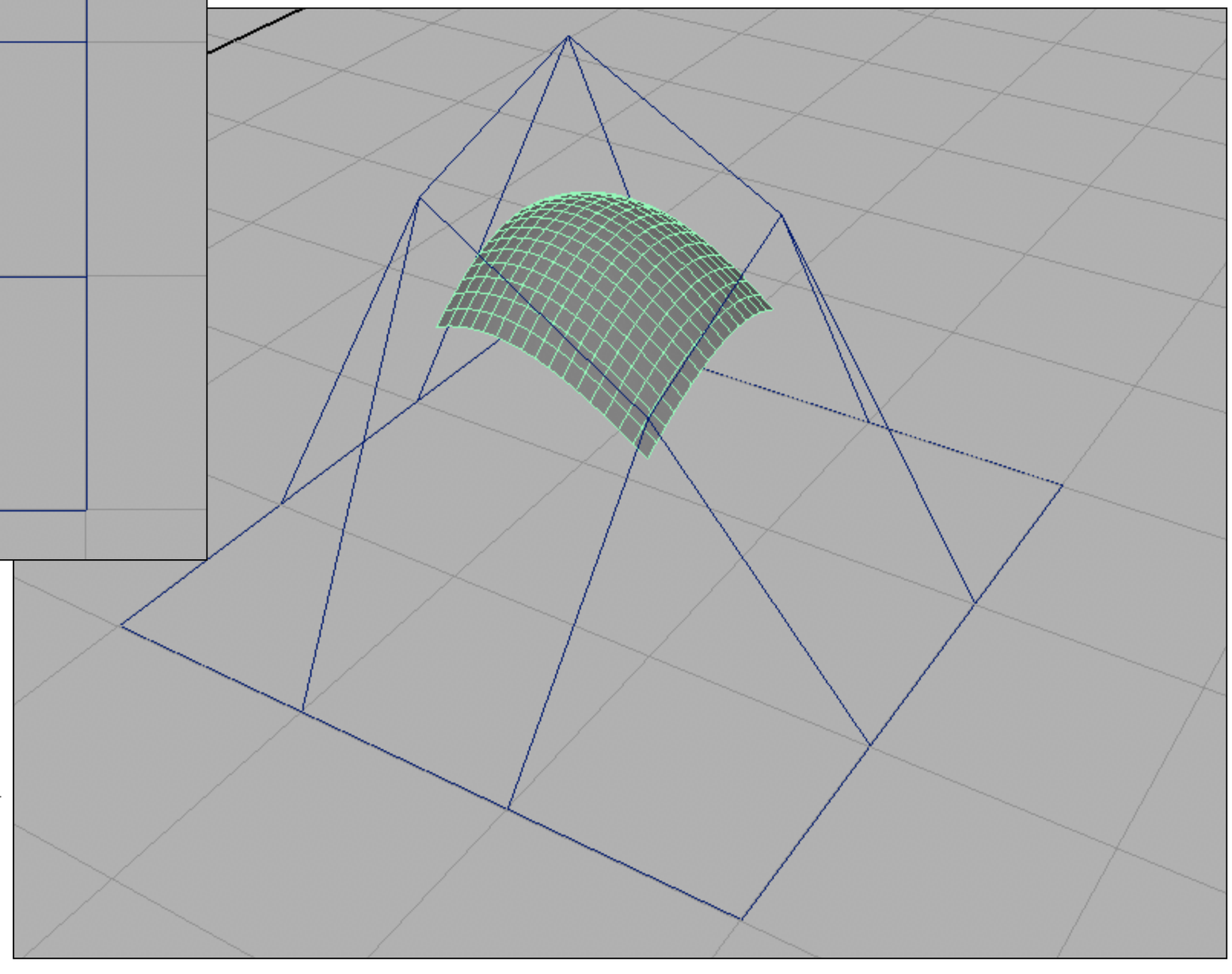
p old "vertex point"

These rules are B-Spline subdivision.

Regular B-Spline Subdivision

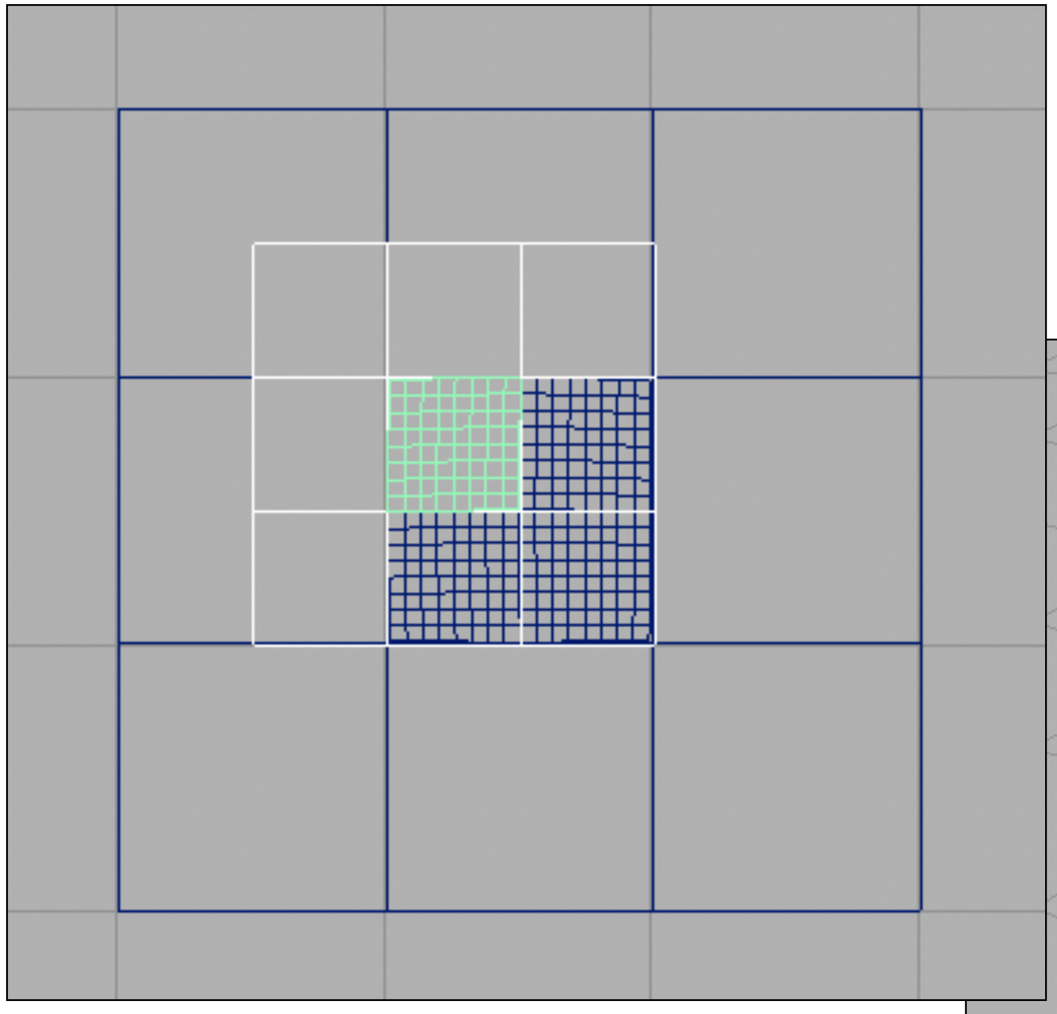


← Orthographic top-down view

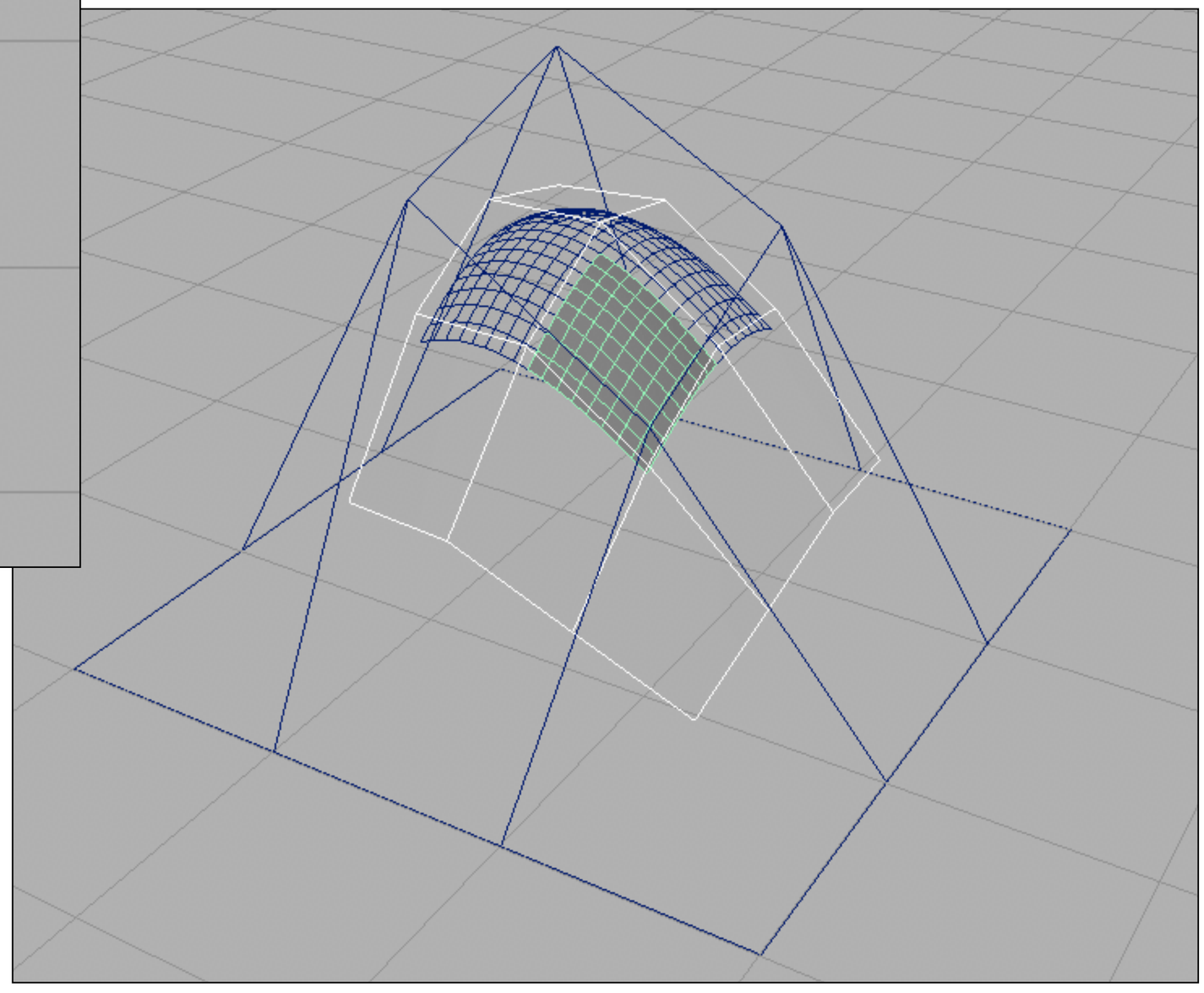


3D Perspective view →

Regular B-Spline Subdivision

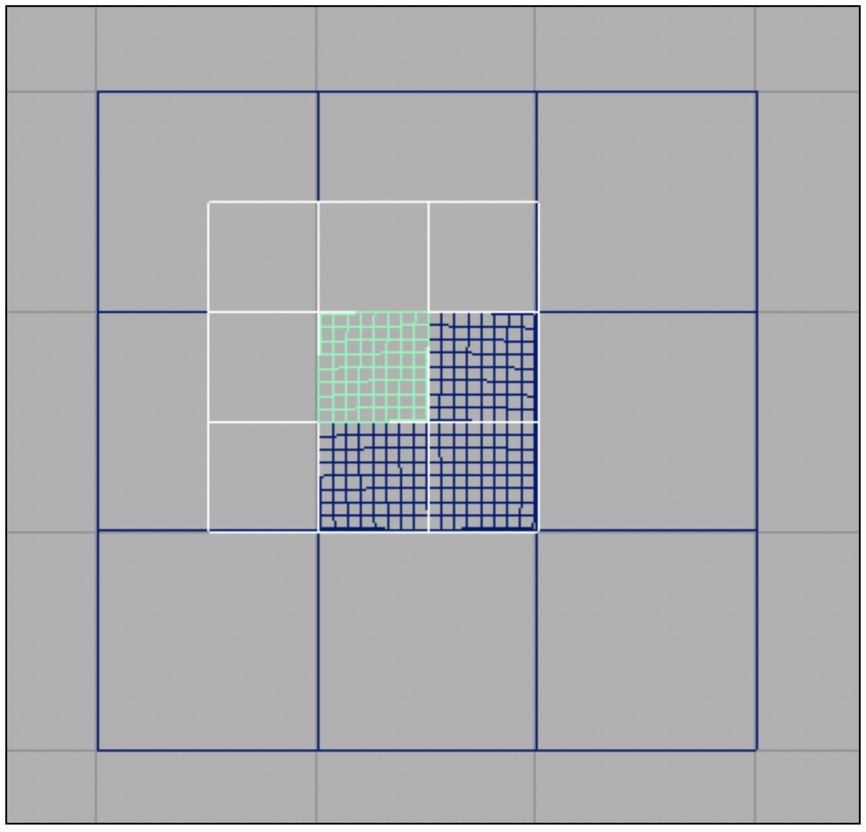


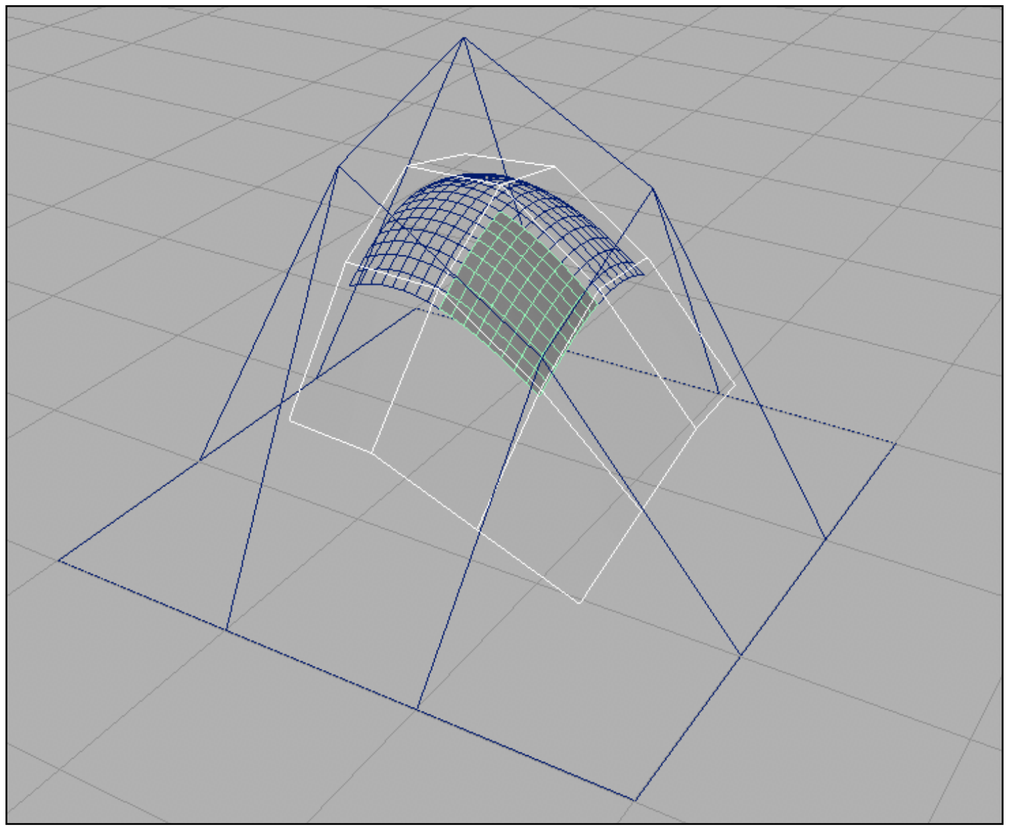
← Orthographic top-down view



3D Perspective view →

Regular B-Spline Subdivision



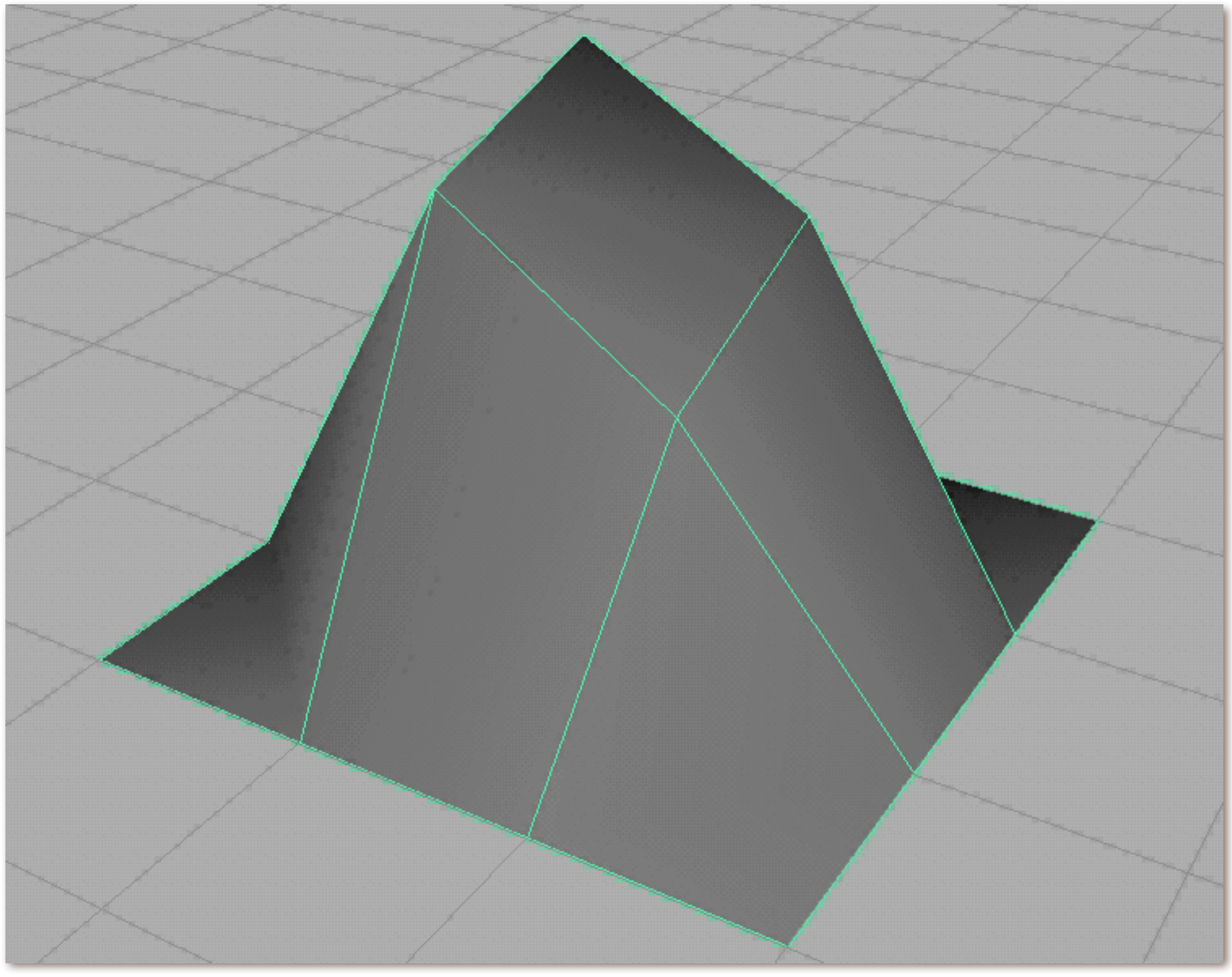


$$\mathbf{x}(u,v) = [1, u, u^2, u^3]\beta_B \, \mathbf{P} \, \beta_B^T [1, v, v^2, v^3]^T$$

$$\mathbf{P}_{11} = \mathbf{H}_{B1} \mathbf{P} \mathbf{H}_{B1}^T$$

Same setup as Bézier, different H matrices.

Regular B-Spline Subdivision



Regular B-Spline Subdivision

